Ans. to the ques. no — 03

There are several difference between Lyophilic sols and Lyophobic sols. These are given below—

Lyophilic sols	Lyophobic sols
1. Prepared by direct mixing with dispersion medium.	1. Not prepared by direct mixing with the medium.
2. Little or no charge on particles	2. Particles carry positive or negative charge.
3. Particles migrate to anode or cathode, or not at all.	3. Particles migrate to either anode or cathode.
4. Precipitated by high concentration of electrolytes.	4. Precipitated by low concentration of electrolytes.
5. Viscosity higher than dispersion medium, set to a gel.	5. Viscosity almost the same as of medium; do not set to a gel.

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Lyophilic sols	Lyophobic sols
6. Particles generally solvated.	6. No solvation of particles.
7. Reversible	7. Irreversible
8. Do not exhibit Tyndall effect.	8. Exhibit Tyndall effect.
9. More stable	9. Less stable
10. Surface tension is lower than dispersion medium	10. Surface tension is nearly same as the dispersion medium.
11. Less scattering.	11. More scattering
12. Mostly of organic nature.	12. Mostly of inorganic nature

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Ans. to the ques. no – 01

i) Colloids :- When the size of the particles of a substance dispersed in a solvent ranges from 1 nm to 200 nm, the system is termed a colloidal solution or simply a colloid.

ii) True solution :- When the size of the particles in the range of 0.1 nm to 1 nm is called true solution.

iii) Suspension :- When the particles size greater than 200 nm, then the solution is called suspension.

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Ans. to the ques. no — 02

Lyophobic sols have to be prepared by special methods. Dispersion method is one of them.

Dispersion method to prepare sols :-

In this method larger macro-sized particles are broken down to colloidal size. In these medium methods, material in bulk is dispersed in another medium.

1) Mechanical dispersion using colloidal mill :- the solid along with the liquid dispersion medium is fed into

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a colloidal mill. The mill consists of two steel plates nearly touching each other and rotating in opposite directions with high speed. The colloid solid particles are ground down to colloidal size and are then dispersed in the liquid to give the sol. colloidal graphite (a lubricant) and printing inks are made by this method.

(2) Bredig's Arc method :- It is used preparing hydrosols of metals ie. Ag, Au and Pt. An arc is struck between

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the two metal electrodes held close together beneath de-ionized water. The water is kept cold by immersing the container in ice/water bath and a trace a alkali (KOH) is added.

3) By peptization :— Peptization is the reverse of coagulation of a sol. The dispersal of a precipitated material into colloidal solution by the action of an electrolyte in solution, is termed peptization. The electrolyte used is called a peptizing agent. Example of preparation of sols by peptization :—

1) Agel where HCl is adding (4—

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is a common ion)

ii) $FeCl_3$ when $Fe(OH)_3$ is adding. Here Fe^{3+} being common ion.

Ans. to the question no — 4

The properties of colloids: There are some properties of colloids such as i) optical properties, ii) kinetic properties and iii) Electrical properties.

The optical properties :—

⇒ when a strong beam of light is passed through a sol and viewed at right angles, the path of ligth shows

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up as a hazy beam or cone. This is due to the fact that sol particles absorb light energy and then emit it in all directions in space. This scattering light as it is called illuminates the path of the beam in the colloidal dispersion. The phenomenon of the scattering of light by the sol particles is called Tyndall effect. The illuminated cone formed by the scattering of light by the sol particles is often referred as Tyndall beam or Tyndall cone. If the sol particles are large enough, the sol may

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be even appear turbid in ordinary light as a result of Tyndall scattering.

We know that True solutions donot show Tyndall effect, since ions or solute molecules are too small to scatter light, the of light passing through a true solution is not visible when viwed from the side. Thus Tyndall effect can be used to distinguish a colloidal solution from a true solution.

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Ans. to the ques. no — 5

Electrical Double Layer :— This theory is involved in electrical properties of colloid as well as sols. This theory describes below —

The surface of colloidal particle acquires a positive charge by selective adsorption of a layer of positive ions around it. This layer attracts counter ions from the medium which form a second layer of negative charges. The combination of two layer of +ve and

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negative charges around the sol particles was called Helmotz double layer. Helmotz thought that positive charges next to the particle surface were fixed, while the layer of negative charges along with the medium were mobile phase.

The two layer i.e. double layer consits that are —

I) A compact layer of positive and negative charges.

II) A Diffuse layer of counter-ions (negative ions) diffused into the medium containing positive ions.

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The combination of the compact and diffuse layer is reffered as the stern double layer after the colloid chemist who first realized its significance. Because of the distribution of charge around the particle, there is a difference in potential between in compact layer and diffuse layer is called Zeta potential.

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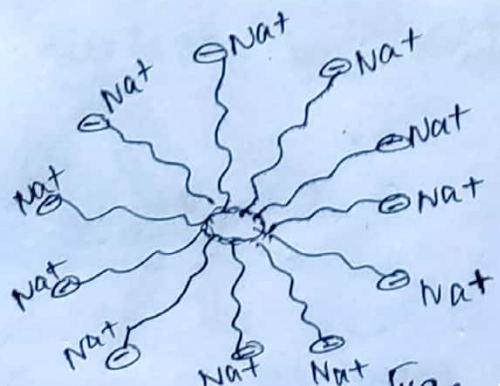
Ans. to the ques. no – 7

Micelles are defined as the colloidal aggregates of soap or detergent molecules formed in the solvent.

Example of micelles :–

Soap or detergents molecule ionizes in water to form an anion & sodium ion. Thus Sodium stearate furnishes stearate anion & sodium cation in aquous solution.

$$C_{17}H_{35}COO^-Na^+ \longrightarrow C_{17}H_{35}COO^- + Na^+$$



$C_{17}H_{35}COO^-$ → Stearate ion

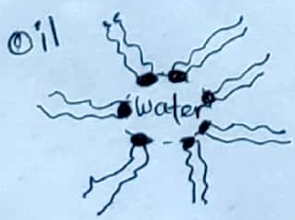
insoluble salt — polar head

Fig: A soap micell

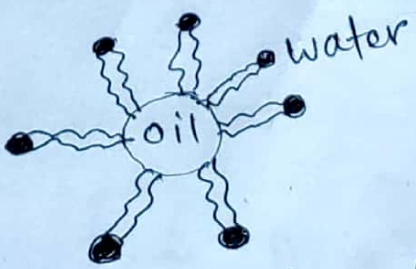
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As many as 70 stearate ions (anions) aggregate to form a micelle of colloidal size. The stearate ion has a long hydrocarbon chain (hydrophobic part) with a polar $-COO^-$ group (hydrophilic part) with at one end. The zig-zag hydrophobic tail is shown by a wavy line and hydrophilic part by a hollow circle.

In the micelle formation, the tails being insoluble in water are directed toward the centre, while the soluble polar heads are on the surface in contact with water.



Reverse Micelle (W/O)



Normal Micelle (O/W)

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Ans. to the ques. no — 6

Gels : A gel is a jelly-like colloidal system in which a liquid is dispersed in a solid medium.

There are two types of gel —

i) Elastic gels and ii) Non-elastic gels.

i) Elastic gels :- Elastic gels are those which posses the property of elasticity. The change their shape on applying force and return to original

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shape when the force is removed.

Examples :— Gelatin, starch and soapes.

ii) Non-elastic gels : Non-elastic gels are those which are rigid i.e. silica gel. These are prepared by appropriate chemical action. Thus silica gel is produced by adding concentrated hydrochloric acid to sodium silicate. Solution of the correct concentration. The resulting molecules of silicic acid polymerize to form

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Silica gel. It has a network linked by covalent bonds which give a strong and rigid structure.

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